Avijit Mondal Shantonu

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Academic Background

Bangladesh University of Engineering and Technology (BUET) *Dhaka, Bangladesh*
Bachelor of Science in Mechanical Engineering *January 2022 – June 2026*

- CGPA: **3.24/4.00**

Govt. Shahpur Madhugram Colege *Khulna, Bangladesh*
Higher Secondary Certificate in Science *2018 – 2020*

- GPA: **5.00/5.00**

Research Interest

Computational Fluid Dynamics (CFD), Multiphase Flow, Droplet Dynamics, Heat Transfer and Thermal Management, Porous Media Flow, Magnetohydrodynamics (MHD), and Renewable Energy Systems.

Relevant Coursework

- Fluid Mechanics
- Thermodynamics
- Heat Transfer
- Numerical Analysis
- Thermo-Fluid System Design
- Mechanics of Materials
- Energy and Environment
- Nuclear Engineering

Under-Graduate Thesis

Effects of Weber Number and Film Thickness on Droplet Dynamics on Thin Liquid Films: Numerical Investigation and Neural ODEs - Based Prediction

Macro-to-Micro Scale Fluids Engineering Lab (MμFEL), Department of Mechanical Engineering, BUET

Supervisor: A.B.M. Toufique Hasan, PhD
Research Focus: CFD, Multiphase Modeling, Droplet Dynamics, FVM and VOF, ML surrogate Modeling
Synopsis:

- Developed a transient 2D axisymmetric multiphase CFD model using FVM–VOF in ANSYS Fluent for single water-droplet impact on thin liquid films.
- Incorporated surface tension and Adaptive Mesh Refinement (AMR) to accurately capture the evolving liquid–air interface.
- Investigated the effects of Weber number and dimensionless film thickness on crown evolution, cavity dynamics, pressure distribution, and velocity fields.
- Explored an ANODE-based predictive model for late-stage crown radius and height, integrating multiphase CFD simulations with data-driven modeling.

Project Experience

Energy and Environment Project *Level 4, Term 2*
A Cost-Effective Standalone Hybrid Renewable Microgrid for Island Electrification Case Study: Saint Martin's Island, Bangladesh

- Conducted techno-economic optimization of a standalone hybrid renewable microgrid for Saint Martin's Island using HOMER Pro, evaluating PV, wind, diesel generator, and battery configurations.
- Identified an optimal PV–Wind–Diesel–Battery configuration with a 99.8% renewable energy fraction and \$0.228/kWh LCOE, demonstrating a cost-effective approach for off-grid island electrification.

Thermo-Fluid System Design *Level 3, Term 2*
Portable Cooling Pad for Laptop

- Designed an R-134a vapor-compression cooling pad to maintain optimal laptop operating temperatures.
- Modeled thermal dynamics in Python, optimizing copper coil heat exchangers (150 W heat duty, $COP = 3$) using the Dittus-Boelter equation.
- Engineered and validated a physical prototype under 200 psi hydraulic pressure testing and flow testing at BUET Laboratories.

Electro-Mechanical System Design and Practice *Level 3, Term 1*
Precision Planter Co-Bot: A Cost-Effective Rocker Boogie Based Robot for Precision Seeding and Fertilizing

- Designed a terrain-adaptive agricultural robot using a rocker-bogie mechanism, PVC chassis, and 3D-printed seed droppers.

- Developed a wheel-speed-synchronized seeding mechanism and performed torque estimation for reliable operation.
- Integrated an Arduino Mega 2560 control system and validated obstacle navigation and precision seed and fertilizer placement.

Ongoing Research

Numerical Investigation of Liquid Droplet Impingement on a Heated Surface

- Developing a transient multiphase CFD model using the VOF method to simulate droplet impact, evaporation, and vapor formation on heated surfaces.
- Incorporating the energy equation and mass transfer to investigate coupled heat transfer and evaporation phenomena.
- Investigating the effects of surface temperature and impact conditions on droplet dynamics, evaporation, and cooling performance.

Publications

1. **Avijit Mondal Shantonu**, A. B. M. Toufique Hasan, "Crown Evolution and Transient Pressure Loading during Droplet Impact on Thin Liquid Films," *Manuscript submitted for review*, 2026.

2. **Avijit Mondal Shantonu**, A. B. M. Toufique Hasan, "Aug.NODE Surrogate for Crown Rim Radius Prediction in Droplet Impact on Thin Liquid Film: A Machine Learning Study,"
Manuscript submitted for review, 2026.

3. Juthi Hira, **Avijit Mondal Shantonu**, Md. Sheikh Fahim-Bin-Salam, Sabid Jubayer Alif, "Precision Planter Co-Bot: A Cost-Effective Rocker-Bogie Based Robot for Precision Seeding and Fertilizing," *2025 4th International Conference on Automation, Robotics and Computer Engineering (ICARCE 2025)*, **Accepted after peer review**, 2025.

Professional Experience

Ashuganj Power Station Company Limited (APSCL) *Brahmanbaria, Bangladesh*
Mechanical Engineering Intern — Power Plant Operation and Maintenance *March 2025*

COMSOL Multiphysics Simulation of Thermofluidic Problems *DCE, BUET*
Completed both Basic and Advanced short courses focusing on thermo-fluidic modeling and numerical analysis. *September 2024*

Technical Skills

- **CAD & Engineering Design:** SolidWorks, AutoCAD, Mechanical System Design
- **Numerical Simulation:** ANSYS Fluent, ANSYS Workbench, COMSOL Multiphysics,
- **Programming & Computational Tools:** MATLAB, Python, C++, LaTeX
- **Microsoft Office Suite:** MS Word, Excel, PowerPoint
- **Research:** Literature Review, Technical Writing, Manuscript Preparation, Research Data Analysis

Awards and Achievements

- **Winner – Assemblaze, Autofest 2024**, Team Pixhawk: Secured 1st place in the competition organised by BUET Automobile Club.
- **Best Volunteer – BUET Admission Test 2023**: Awarded a Certificate of Appreciation for outstanding service as a Rover Scout volunteer during the undergraduate admission examination.

Extracurricular Activities

- **Bangladesh Scouts** — Member (Scout ID: CD7164) *February 2022 – Present*
- **BUET Rover Scout Society** — Vice-President *March 2025 – June 2026*
- **Badhon, BUET Zone** — Volunteer *February 2022 – Present*
  Volunteer with a student-led blood donation and social service organization, supporting blood donation and community service activities.

References

Dr. A. B. M. Toufique Hasan
Professor, Mechanical Engineering
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Dr. Abdul Aziz Shuvo
Assistant Professor, Mechanical Engineering
BUET | shuvo@me.buet.ac.bd